V12 Engine & Tracktor

Technical Documentation

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January 7, 2026

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1 Introduction

This document describes the architecture, design philosophy, and systems behind the **V12 Engine** and the game **Tracktor**.

V12 is a real-time, networked 3D engine supporting both desktop and VR interaction paradigms. Tracktor is a flagship project built on top of V12 to validate engine features and workflows.

This documentation targets:

- Engine developers
- Game developers
- Tooling and editor contributors

2 V12 Engine Overview

V12 is a modular, ECS-driven engine designed for:

- Real-time collaborative scenes
- Desktop and VR parity
- Network-first architecture
- Extensible component behavior

2.1 Core Features

- Entity Component System (ECS)
- Custom Components with Addons
- Scene synchronization
- Event-driven architecture
- Integrated inspector and tooling

3 Entity Component System

V12 uses a data-oriented ECS design.

3.1 Entities

Entities are lightweight identifiers with no inherent behavior.

3.2 Components

Components store structured data attached to entities.

3.3 Addons

Addons are sub-components that modify component behavior at runtime.

- Addons may inject logic
- Addons may modify synchronization rules
- Addons may expose editor tooling

4 VR and Desktop Support

V12 supports both VR and desktop users within the same scene.

4.1 Input Abstraction

All input devices map into a unified interaction layer:

- Mouse / Keyboard
- VR Controllers
- Hand Tracking

4.2 Hands and Avatars

Hands are first-class scene objects and participate in:

- Physics
- Interaction events
- Network replication

5 Networking and Scene Sync

Scenes are synchronized across peers using a deterministic event model.

5.1 Scene Syncer

The scene syncer handles:

- Entity creation and destruction
- Component replication
- Authority resolution

5.2 Events

Events are serialized and replayed to maintain consistency.

6 Editor and Tooling

V12 includes built-in tools for inspection and debugging.

6.1 Scene Inspector

The scene inspector allows live modification of:

- Entities
- Components
- Addons

7 Tracktor Game Overview

Tracktor is a collaborative 3D experience built to stress-test V12.

7.1 Design Goals

- Persistent shared spaces
- Live interaction
- Modifiable runtime content

8 Future Work

Planned areas of development include:

- Improved scripting interfaces
- Expanded addon system
- Performance optimizations
- Public modding APIs